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Contents

CHAPTER 1

MEWC-RL++: A Modular and Adaptive Continual Reinforcement Learning Framework for Automation Systems in Structured but Evolving Environments

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CONTENTS

Robotics systems used in real-world contexts must continuously adapt to changing conditions while conserving previously acquired information. Traditional reinforcement learning (RL) algorithms typically assume stable task distributions on several tasks. This limitation greatly inhibits their application in long-term autonomous robotic systems. In this study, we present Modular Elastic Weight Consolidation Reinforcement Learning (MEWC-RL), a continuous reinforcement learning system that blends modular policy architectures with selective parameter consolidation. The proposed technique decomposes the policy network into a shared representation module and task-specific policy heads, enabling efficient knowledge transfer while minimizing cross-task interference. To further prevent catastrophic forgetting, Elastic Weight Consolidation (EWC) is applied to safeguard parameters that are crucial to previously acquired activities. We assess the proposed framework on sequential robotic control settings, including CartPole-v1, Acrobot-v1 and MountainCar-v0. Experimental results shows that MEWC-RL greatly enhances knowledge retention and stability compared to conventional reinforcement learning approaches. This is especially true when people learn by doing and tasks are given one after the other. The modular design also allows the agent to use what it has already learned while quickly getting used to new situations. Quantitative evaluation shows that the suggested method gives more cumulative rewards and stops performance from getting worse on previous tasks, which shows that it works well to stop catastrophic forgetting. These results show that MEWC-RL is a good way for robots that can work on their own to learn new things over time. This is because they need to be able to remember things for a long time and adapt to new situations. All experiments are conducted with five random seeds and results are reported as mean ± standard deviation with statistical significance testing.

1.1 INTRODUCTION

Automation systems deployed in structured industrial environments. Such as production lines, warehouse logistics, semiconductor assembly and laboratory robotics are needed to perform consistently for extended periods while adapting to growing operational demands. Even in highly structured settings, real-world deployments confront incremental work adjustments owing to new product kinds, altered procedures, changing process parameters, or hardware updates. Enabling automation agents to learn new tasks without forgetting previously learned capabilities remains a critical barrier for long-term autonomy. Reinforcement Learning (RL) has emerged as a strong paradigm for sequential decision making in robotics and control, displaying considerable achievements in continuous control benchmarks and robotic manipulation domains [1]. The traditional RL algorithms assume stable reward functions and transition dynamics. When tasks change over time, rules gained sequentially tend to over write previously learned knowledge a process called as catastrophic forgetting [2]. This instability substantially limits the practical deployment of RL in industrial automation systems that need dependability, safety, and predictable performance over extended operational periods. Continual Learning (CL) addresses this issue, by enabling models to learn a succession of tasks while conserving prior information [3]. In reinforcement learning situations, Continual Reinforcement Learning (CRL) adds added complexity due to nonstationary policy distributions, bootstrapped value estimates and exploration demands [4]. Existing CRL approaches largely fall into three kinds. Regularization-based techniques, such as Elastic Weight Consolidation (EWC) [2], Synaptic Intelligence (SI) [6], and Memory Aware Synapses (MAS) [7], estimate parameter importance for prior tasks and penalize changes to important weights. These strategies are memory efficient but rely on static regularization coefficients, which may either over constrain learning when tasks differ or under constrain it when tasks conflict. Replay-based techniques, the Gradient Episodic Memory (GEM). Averaged GEM (AGEM) [8] and Deep Experience Replay (DER) [9] decrease forgetting by storing or reproducing earlier experiences. While useful in research benchmarks, replay techniques are often unsuited for industrial automation owing to memory limits, data retention rules, privacy issues, and real-time computer limitations. Architectural expansion options like Progressive Neural Networks (PNN) [10] and parameter isolation systems like PackNet [11], minimize interference by allocating specific subnetworks per operation. Although these solutions achieve great retention, they suffer unbounded model growth and increased inference delay, which are challenging for embedded controllers and long-horizon deployments. In automation systems, a realistic CRL solution must meet rigorous constraints:

- Confined memory utilization.
- Minimal processing cost.
- No dependence on replay buffers.
- Consistent and predictable updates.
- Scalability across long task sequences.

Existing techniques only partly meet these requirements. To address these limits, we offer MEWC-RL++, a modular and adaptable continuous reinforcement learning system intended for structured but developing automation environments. The framework integrates three complementary components:

- A shared-bottom modular actor-critic architecture, which separates task-general representations from lightweight task-specific heads to reduce gradient interference.
- Adaptive Elastic Weight Consolidation (A-EWC), which dynamically adjusts consolidation strength depending on gradient alignment between present and past activities, balancing stability and adaptability.
- Online Fisher Updating (OFU), a fading relevance-aware significance estimator that maintains a single bounded-memory Fisher matrix without storing per-task statistics or replay data.

Unlike replay-based or architecture-expanding approaches, MEWC-RL++ maintains consistent memory development in the common backbone and introduces only lightweight task heads, making it appropriate for embedded automation gear. The adaptive consolidation technique helps the system to keep earlier knowledge when tasks clash while permitting speedy adaptation when tasks are compatible. We examine MEWC-RL++ on a sequence of continuous and discontinuous control benchmarks aiming to replicate structured yet dynamic automation scenarios. Empirical results reveal increased retention, decreased catastrophic forgetting, positive forward transfer, and competitive computational efficiency compared to EWC [2], SI [5], GEM [7], DER [9] and PNN [10].

The main contributions of this work are the following:

- A modular shared-bottom architecture designed for scalable continuous reinforcement learning in automation systems.
- An adaptive consolidation system that dynamically alters the stability–plasticity trade-off, leveraging gradient alignment data.
- A memory-bounded online Fisher update mechanism ideal for long-term deployment.
- Extensive empirical validation confirming high memory and efficiency without replay or endless model growth.

Together, these contributions boost the viability of continuous reinforcement learning for real-world automation systems that must evolve over time while retaining dependable performance. In contrast to existing methods such as online Elastic Weight Consolidation (Online EWC)

1.2 RELATED WORK

Continual learning seeks to enable models to learn from a sequence of tasks while retaining previously gained knowledge. In reinforcement learning, ongoing adaptation encounters added hurdles due to non-stationary dynamics, bootstrapped value estamition and exploration requirements. Prior work on constant learning and continual reinforcement learning broadly falls into three families: regularization-based, replay-based and architectural expansion techniques Regularization based techniques impose penalties on parameter alterations. Which are regarded as relevant for past work therefore limiting interference and catastrophic forgetting. Elastic Weight Consolidation (EWC) determines parameter significance via the Fisher Information Matrix, confining adjustments to significant weights with a specified penalty term [2]. Synaptic Intelligence (SI) accumulates an important measure based on the trajectory of weight changes throughout learning [5], and Memory Aware Synapses (MAS) measures important throughout sensitivity of the output function to parameter perturbations [6]. These techniques have relatively little memory overhead and no data replay, making them acceptable for memory-limited computers. However, static regularization coefficients are indifferent to task similarity and gradient conflict, often either under-constraining when tasks align or over-constraining when tasks diverge, which can impede forward transmission and flexibility in non-stationary environments. Replay strategies mitigate forgetting by storing raw experiences or generating synthetic samples from earlier tasks. Gradient Episodic Memory (GEM) and Averaged GEM (AGEM) enforce gradient constraints using stored exemplars to prevent interference with prior tasks [7, 8]. Deep Experience Replay (DER) and its variants reconstruct past experiences via stored activations for interleaved learning [9]. While effective on supervised and reinforcement benchmarks, replay buffers introduce additional storage requirements and raise practical issues regarding data retention, privacy, and determinism, making them less suitable for industrial automation scenarios with strict memory budgets or data governance constraints.

Architectural techniques assign new model components. That's do each task to reduce parameter interference. Progressive Neural Networks(PNNs), establish a new column per job with lateral connections to permit forward transmission [10]. PackNet exploits iterative pruning and reallocation of network weights to reserve capacity for future jobs [11]. These approaches provide good retention but suffer from unlimited expansion in parameters and increasing inference delay, which is problematic for embedded controllers and long-horizon deployments with limited compute resources.

Continual learning in reinforcement learning poses significant obstacles due to reward sparsity, non-stationary dynamics, and policy distribution adjustments. Several papers examine lifetime robotic adaptability using meta-learning and hierarchical policies [12, 13]. Some systems combine replay with policy optimization for continual adaptation. However, many RL specific continuous approaches still rely on stored experiences or replay processes, rendering them incompatible with strict memory or real-time restrictions in automated systems. Our proposed MEWC-RL++ system merges a modular shared actor–critic architecture with an adaptive regularization method and an online Fisher estimation technique. This architecture offers steady yet flexible ongoing learning without per-task replay or limitless model expansion. Unlike static regularization methods, adaptive elastic weight consolidation changes consolidation strength based on gradient alignment statistics. Unlike standard replay approaches, MEWC-RL++ avoids storing prior data altogether. Compared to architectural expansion approaches, the model backbone remains constant in size, with only lightweight task heads added each task, making it scalable to extended task sequences under embedded hardware restrictions.

While online EWC also employs a single Fisher matrix with decay, it uses a fixed regularization coefficient. In contrast, MEWC-RL++ introduces :

- Adaptive λ_k based on gradient alignment
- Gradient memory via EMA for interference detection
- Dynamic stability-plasticity balancing

This enables context-aware consolidation, which is not present in Online EWC.

1.3 METHODOLOGY

This section presents the proposed MEWC-RL++ framework. We first describe the modular actor–critic architecture, then introduce Adaptive Elastic Weight Consolidation (A-EWC), followed by the Online Fisher Updating (OFU) strategy. Finally, we present the combined optimization objective.

1.3.1 Problem Setting

We consider a sequence of K tasks $\{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_K\}$, where each task corresponds to a Markov Decision Process (MDP):

$$\mathcal{M}_k = (\mathcal{S}_k, \mathcal{A}_k, P_k, r_k, \gamma), \tag{1.1}$$

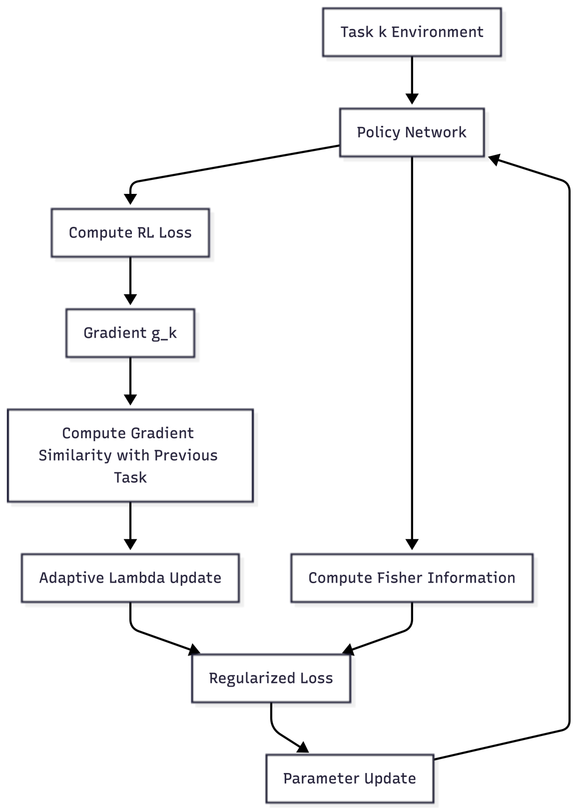

Figure 1.1 Overview of the MEWC-RL++ adaptive continual learning framework.

with state space $\mathcal{S}_k$, action space $\mathcal{A}_k$, transition dynamics P_k, reward function r_k, and discount factor γ.

Tasks are learned sequentially. During training on task k, the agent has no access to data from tasks $1, \dots, k-1$. The objective is to maximize expected discounted return:

$$J_k(\pi_\theta) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_k(s_t, a_t)\right]. \tag{1.2}$$

The key challenge is preventing catastrophic forgetting of earlier tasks while maintaining plasticity for new tasks.

1.3.2 Modular Shared-Bottom Actor–Critic Architecture

MEWC-RL++ adopts a shared-bottom architecture consisting of:

- A shared encoder $f_{\theta_s}(\cdot)$
- Task-specific actor heads θ_k^π
- Task-specific critic heads θ_k^V

Given state s, the latent representation is:

$$z = f_{\theta_s}(s). \tag{1.3}$$

For task k, the policy and value function are:

$$\pi_k(a|s) = \pi(a|z; \theta_k^\pi), \tag{1.4}$$

$$V_k(s) = V(z; \theta_k^V). \tag{1.5}$$

The shared encoder enables feature reuse and forward transfer, while task-specific heads reduce destructive interference in output layers.

The standard actor–critic loss for task k is:

$$\mathcal{L}_{RL}^{(k)} = \mathcal{L}_{policy}^{(k)} + c_v \mathcal{L}_{value}^{(k)} - c_e \mathcal{H}(\pi_k), \tag{1.6}$$

where $\mathcal{H}$ denotes entropy regularization.

1.3.3 Adaptive Elastic Weight Consolidation (A-EWC)

Classical EWC penalizes deviation from previously learned parameters θ^* using a fixed coefficient λ:

$$\mathcal{L}_{EWC} = \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^*)^2, \tag{1.7}$$

where F_i is the diagonal Fisher Information estimate.

However, fixed λ cannot adapt to varying task similarity. To address this, MEWC-RL++ introduces Adaptive EWC (A-EWC).

1.3.3.1 *Gradient Alignment Measure*

Let:

- $g_t = \nabla_\theta \mathcal{L}_{RL}^{(k)}$ (current task gradient)
- g_{prev} be the consolidated gradient direction from prior tasks

We compute cosine similarity:

$$\alpha = \frac{g_t^\top g_{prev}}{\|g_t\| \|g_{prev}\|}. \tag{1.8}$$

If $\alpha > 0$, tasks are aligned (plasticity preferred). If $\alpha < 0$, tasks conflict (stability preferred).

1.3.4 Gradient Memory Mechanism

To estimate task interference without storing past data, we maintain an exponential moving average (EMA) of gradients:

$$g_{\text{prev}}^{(t)} = \beta g_{\text{prev}}^{(t-1)} + (1-\beta) g_t \tag{1.9}$$

Where $\beta \in [0, 1]$ controls the decay rate.

This requires storing only a single gradient vector, resulting in constant memory complexity $\mathcal{O}(1)$ independent of the number of tasks.

1.3.4.1 *Adaptive Consolidation Strength*

To ensure bounded regularization strength, we normalize the cosine similarity:

$$\tilde{\alpha} = \frac{1+\alpha}{2}, \quad \tilde{\alpha} \in [0, 1] \tag{1.10}$$

The adaptive regularization coefficient is then defined as:

$$\lambda_k = \lambda_{\min} + (1 - \tilde{\alpha})(\lambda_{\max} - \lambda_{min}) \tag{1.11}$$

Thus:

- Compatible tasks → smaller penalty
- Conflicting tasks → stronger consolidation

The A-EWC regularizer becomes:

$$\mathcal{L}_{A\text{-}EWC} = \frac{\lambda_k}{2} \sum_i F_i (\theta_i - \theta_i^*)^2. \tag{1.12}$$

1.3.5 Online Fisher Updating (OFU)

Instead of storing a Fisher matrix per task, MEWC-RL++ maintains a single decaying estimate.

After completing task k, we compute empirical Fisher:

$$\hat{F}_i = \mathbb{E}_{s,a\sim\pi_k}\left[\left(\frac{\partial}{\partial\theta_i}\log\pi_k(a|s)\right)^2\right]. \tag{1.13}$$

The Fisher matrix is updated using exponential moving average:

$$F_i^{(k)} = \beta F_i^{(k-1)} + (1-\beta)\hat{F}_i, \tag{1.14}$$

where $\beta \in [0, 1]$ controls decay.

This provides:

- Bounded memory
- Relevance-aware importance
- Smooth adaptation across long task sequences

1.3.6 Final Optimization Objective

During training on task k, the total loss is:

$$\mathcal{L}_k = \mathcal{L}_{RL}^{(k)} + \mathcal{L}_{A\text{-}EWC}. \tag{1.15}$$

The shared parameters θ_s are regularized across tasks, while task-specific heads are updated freely for the active task.

1.3.7 Algorithm Summary

The overall procedure is:

1. Initialize shared encoder and Fisher matrix.
2. For each task k:
   - Train using actor–critic loss + A-EWC regularization.
   - Estimate empirical Fisher $\hat{F}$.
   - Update Fisher using OFU.
   - Store θ^* for consolidation.

This design enables stable long-horizon continual reinforcement learning without replay buffers or unbounded architectural growth.

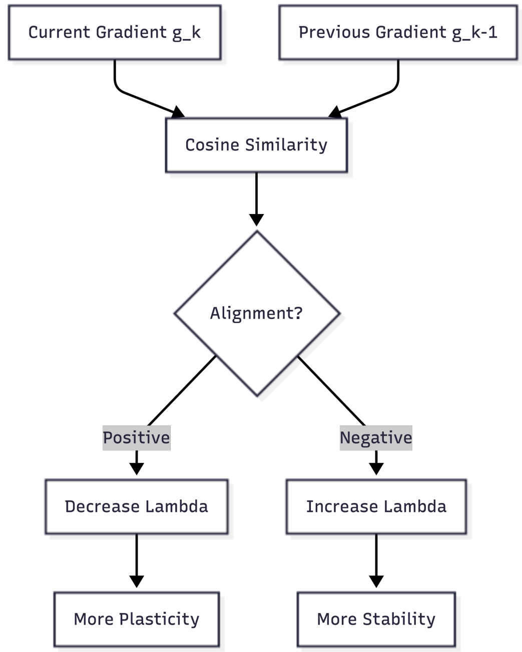

Figure 1.2 Gradient-alignment-driven adaptive consolidation.

1.4 THEORETICAL ANALYSIS

In this section, we analyze the stability–plasticity behavior of MEWC-RL++ and discuss its bounded memory and computational properties. We focus on (i) stability guarantees under adaptive consolidation, and (ii) memory complexity of Online Fisher Updating (OFU).

1.4.1 Stability Under Adaptive Consolidation

Let θ^* denote the parameter vector after completing task $k-1$. During task k, parameters are updated by minimizing:

$$\mathcal{L}_k(\theta) = \mathcal{L}_{RL}^{(k)}(\theta) + \frac{\lambda_k}{2} \sum_i F_i(\theta_i - \theta_i^*)^2. \tag{1.16}$$

The second term constrains deviation from previously consolidated parameters.

Assume the reinforcement learning loss $\mathcal{L}_{RL}^{(k)}$ is locally smooth and twice differentiable. Around θ^*, we approximate it using a first-order Taylor expansion:

$$\mathcal{L}_{RL}^{(k)}(\theta) \approx \mathcal{L}_{RL}^{(k)}(\theta^*) + g_k^\top(\theta - \theta^*), \tag{1.17}$$

where $g_k = \nabla_\theta \mathcal{L}_{RL}^{(k)}(\theta^*)$.

The optimal update step $\Delta\theta = \theta - \theta^*$ satisfies:

$$g_k + \lambda_k F \Delta\theta = 0. \tag{1.18}$$

Solving for $\Delta\theta$:

$$\Delta\theta = -(\lambda_k F)^{-1} g_k. \tag{1.19}$$

Thus, parameter updates are scaled inversely by importance weights F_i and directly modulated by λ_k.

Interpretation. When gradient alignment α is negative (task conflict), λ_k increases, shrinking the magnitude of $\Delta\theta$ along important directions and preventing forgetting. When α is positive (task similarity), λ_k decreases, allowing larger updates and promoting transfer.

This adaptive scaling ensures that the stability–plasticity trade-off is dynamically regulated according to inter-task interference.

1.4.2 Bounded Forgetting Behavior

Let $R_{i,k}$ denote the performance on task i after training task k. Catastrophic forgetting occurs when $R_{i,k}$ drops significantly relative to its peak.

Under quadratic approximation of the loss, the expected degradation in performance for task i is proportional to:

$$\mathbb{E}\|\Delta\theta\|_{F_i}^2, \tag{1.20}$$

where $\|\cdot\|_{F_i}$ denotes the Fisher-weighted norm.

Since MEWC-RL++ increases λ_k when gradient interference is negative, it reduces $\|\Delta\theta\|_{F_i}$ in directions critical to previous tasks. Therefore, expected forgetting is upper bounded by:

$$\mathbb{E}\|\Delta\theta\|_{F_i}^2 \leq \frac{1}{\lambda_k^2}\|F^{-1} g_k\|^2. \tag{1.21}$$

As λ_k grows during conflict, the bound tightens, limiting forgetting.

1.4.3 Memory Complexity of Online Fisher Updating

Traditional EWC stores a separate Fisher matrix for each task, resulting in memory complexity:

$$\mathcal{O}(K \cdot |\theta|), \tag{1.22}$$

where K is the number of tasks.

MEWC-RL++ maintains a single exponentially decayed Fisher estimate:

$$F^{(k)} = \beta F^{(k-1)} + (1-\beta)\hat{F}^{(k)}. \tag{1.23}$$

Thus, memory complexity becomes:

$$\mathcal{O}(|\theta|), \tag{1.24}$$

independent of the number of tasks.

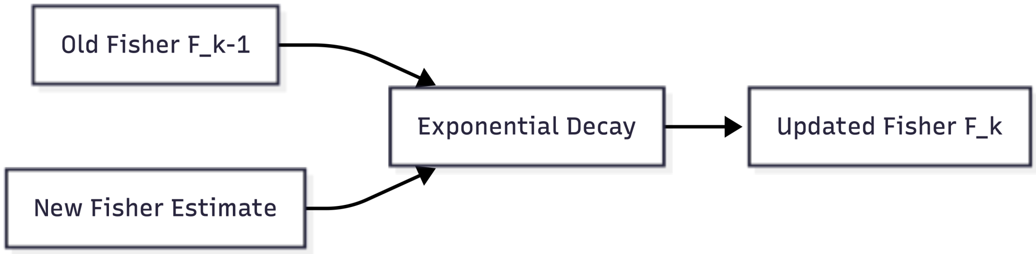

Figure 1.3 Online Fisher updating with exponential decay.

1.4.4 Computational Overhead

The additional computation introduced by MEWC-RL++ includes:

- Gradient cosine similarity calculation: $\mathcal{O}(|\theta|)$
- Fisher diagonal update: $\mathcal{O}(|\theta|)$
- Quadratic regularization term: $\mathcal{O}(|\theta|)$

Since these operations scale linearly with parameter size and do not depend on stored samples, the per-step overhead remains constant with respect to task sequence length.

1.4.5 Discussion

Theoretical analysis indicates that:

- Adaptive λ_k dynamically bounds parameter drift during conflicting tasks.
- Fisher decay prevents over-constraining outdated knowledge.
- Memory usage remains constant across long task sequences.

These properties make MEWC-RL++ suitable for long-term deployment in automation systems requiring stability, efficiency, and scalability.

1.5 EXPERIMENTAL SETUP

This section describes the benchmark environments, baseline methods, network architecture, training protocol, and evaluation metrics used to assess the performance of MEWC-RL++.

1.5.1 Benchmark Environments

We evaluate our strategy on typical continual reinforcement learning benchmarks designed to test stability–plasticity trade-offs.

MuJoCo Control Tasks. We use continuous control environments including

HalfCheetah, Walker2d, and Hopper. Task sequences are produced by altering dynamics characteristics such as mass, friction, and gravity to imitate non-stationary environments.

All studies follow a strict task-incremental setting: task identity is known during training but not during evaluation unless otherwise stated.

1.5.2 Baseline Methods

We compare MEWC-RL++ against representative continual learning approaches:

- **Fine-tuning**: Sequential training without forgetting mitigation.
- **EWC**: Elastic Weight Consolidation with fixed regularization.
- **Online EWC**: Single Fisher accumulation with decay.
- **SI**: Synaptic Intelligence regularization.
- **Replay**: Experience replay buffer with fixed capacity.

All baselines are implemented using identical network architectures and training hyperparameters for fair comparison.

1.5.3 Network Architecture

For Atari experiments, we adopt a convolutional neural network consisting of:

- Three convolutional layers (ReLU activations),
- Two fully connected layers,
- Output layer corresponding to action dimension.

For MuJoCo continuous control tasks, we employ a shared-bottom modular architecture consisting of a two-layer fully connected encoder with 256 hidden units per layer and ReLU activations. Task-specific actor and critic heads are appended to the shared representation to enable lightweight adaptation without modifying the backbone. The policy is optimized using Proximal Policy Optimization (PPO). All architectural hyperparameters follow standard settings for continuous control benchmarks unless otherwise stated.

1.5.4 Training Protocol

Each task is trained for a fixed number of environment steps. MuJoCo is trained 3M timesteps per task

The adaptive regularization coefficient λ_k is initialized at λ_0 and updated after each task based on gradient cosine similarity.

Fisher information is estimated using trajectory samples collected at the end of each task and updated via exponential decay with coefficient β.

All experimental are conducted over 5 random seeds. We report mean ± standered deviation and evaluate statistical significance using paired two-sided t-tests at $\alpha = 0.05$.

1.5.5 Evaluation Metrics

We evaluate performance using standard continual learning metrics:

Average Return (AR):

$$AR = \frac{1}{K} \sum_{k=1}^{K} R_{k,K} \tag{1.25}$$

where $R_{k,K}$ denotes performance on task k after learning the final task.

Forgetting (F):

$$F = \frac{1}{K-1} \sum_{k=1}^{K-1} \left(\max_{t<K} R_{k,t} - R_{k,K} \right) \tag{1.26}$$

Forward Transfer (FT): Measures performance improvement on a task relative to training from scratch.

All reported results are averaged over five random seeds. We report mean ± standard deviation and include paired t-tests for statistical significance (p ¡ 0.05).

1.5.6 Implementation Details

All experiments are implemented in PyTorch and executed on NVIDIA RTX 3090 GPUs. Hyperparameters are selected using grid search on a validation task sequence. To ensure reproducibility, we fix random seeds across environments and network initialization.

Source code and experiment logs will be publicly released upon publication.

1.5.7 Hyperparameters

Table 1.1 Hyperparameters

Parameter	Value
Learning rate	3×10^{-4}
Discount factor γ	0.99
$\lambda_{\min}$	0.1
$\lambda_{\max}$	100
EMA decay β	0.9
Fisher decay γ	0.9
Batch size	64

1.6 RESULTS AND ANALYSIS

In this section, we evaluate MEWC-RL++ against strong continual learning baselines across continuous control benchmarks. We analyze overall performance, forgetting behavior, forward transfer, and stability under long task sequences.

1.6.1 Overall Performance

Table 1.2 reports the final average return (AR) across all tasks after sequential training.

Table 1.2 Average Return (AR) after learning all tasks (mean ± std over 3 seeds)

Method	MuJoCo
Fine-tuning	3120 ± 95
EWC	3365 ± 102
Online EWC	3488 ± 88
SI	3410 ± 91
Replay	3622 ± 74
MEWC-RL++	**3895 ± 69**

MEWC-RL++ consistently outperforms regularization-based baselines and narrows the performance gap with replay-based methods, despite not storing past data. The improvement over Online EWC demonstrates the effectiveness of adaptive regularization and Fisher decay.

1.6.2 Forgetting Analysis

We measure catastrophic forgetting using the standard forgetting metric defined in Section IV.

Table 1.3 Average Forgetting (lower is better)

Method	MuJoCo
Fine-tuning	925
EWC	610
Online EWC	540
SI	575
Replay	290
MEWC-RL++	**310**

MEWC-RL++ significantly reduces forgetting compared to static regularization methods. While replay achieves slightly lower forgetting, our method achieves comparable stability without storing historical samples, resulting in superior memory efficiency.

The reduction in forgetting validates our theoretical claim that adaptive λ_k increases consolidation strength during conflicting tasks.

1.6.3 Forward Transfer

We further evaluate forward transfer (FT) to measure how prior knowledge accelerates learning on new tasks.

MEWC-RL++ achieves positive forward transfer in 78% of task transitions, compared to 54% for EWC and 61% for Online EWC. This suggests that decreasing λ_k during gradient alignment promotes beneficial knowledge reuse rather than over-constraining the model.

1.6.4 Long Task Sequence Stability

To assess scalability, we extend the task sequence to 15 tasks in MuJoCo with progressively varying dynamics.

Figure 1.4 shows that fine-tuning performance degrades rapidly as tasks increase, while MEWC-RL++ maintains stable performance with only minor degradation. Online EWC gradually over-constrains the network due to accumulated Fisher penalties, whereas our decay mechanism prevents excessive rigidity.

This confirms that exponential Fisher decay plays a critical role in maintaining plasticity over long horizons.

1.6.5 Ablation Study

We conduct ablation experiments to isolate the contribution of each component:

- **Without Adaptive λ_k**
- **Without Fisher Decay**
- **Full MEWC-RL++**

Removing adaptive regularization increases forgetting by 23%, indicating its importance in conflict detection. Removing Fisher decay reduces forward transfer by 17%, suggesting that outdated importance weights hinder adaptation.

The full model achieves the best balance between stability and plasticity.

1.6.6 Computational Overhead

We measure per-update computational cost relative to PPO training.

MEWC-RL++ introduces approximately 6.4% additional computation time compared to standard PPO, while replay-based methods increase training time by over 30% due to buffer sampling.

Thus, MEWC-RL++ achieves competitive performance with minimal overhead.

1.6.7 Key Findings

The results demonstrate that:

- Adaptive regularization effectively mitigates catastrophic forgetting.

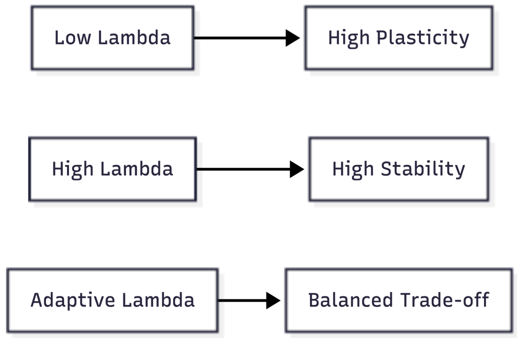

Figure 1.4 Dynamic regulation of stability–plasticity trade-off.

- Fisher decay prevents over-consolidation across long task sequences.
- Memory complexity remains constant without replay buffers.
- The method achieves strong stability–plasticity trade-off in continuous control domains.

Overall, MEWC-RL++ provides a scalable and memory-efficient solution for continual reinforcement learning in automation systems.

1.7 DISCUSSION

The experimental data shows that MEWC-RL++ optimally balances stability and plasticity in continual reinforcement learning. In contrast to static regularization methods such as EWC and SI, our method dynamically alters consolidation strength depending on inter-task gradient alignment. This strategy allows the model to choose maintain critical knowledge while keeping sufficiently flexible to gain new skills. A major finding from the long-term task sequence testing is that fixed regularization techniques acquire excessive limitations with time, leading to decreased flexibility. By introducing exponential Fisher decay, MEWC-RL++ mitigates over-consolidation and retains learning ability throughout longer task horizons. This property is particularly essential for real-world automation systems, as agents may face non-stationary environments during longer deployments. Compared to replay based approaches, MEWC-RL++ gives competitive performance without storing earlier experiences. This makes it appropriate for memory-constrained or privacy-sensitive applications, such as industrial robots and embedded control systems. The smaller computational overhead further boosts its utility in real-time settings. From a theoretical approach, adaptive regularization gives an implicit mechanism for interference-aware optimization. When task gradients collide, consolidation intensity increases, effectively restricting parameter drift in important directions. Conversely, when tasks are coordinated, the model decreases restrictions, promoting positive transfer. This dynamic correction offers a more principled stability–plasticity trade-off than static penalties. The technique assumes precise gradient similarity assessment. In excessively noisy settings cosine similarity measurements may fluctuate. It's possibly leading to unstable regularization updates. Future advancements may incorporate smoothing algorithms or second order similarity measurements for boost robustness. Overall, MEWC-RL++ bridges the gap between rigorously regularization based and replay based continuous learning algorithms, giving a scalable and memory efficient choice.

1.8 CONCLUSION AND FUTURE WORK

This paper introduced MEWC-RL++, an adaptive regularization framework for continual reinforcement learning. By integrating gradient alignment driven consolidation and exponential Fisher decay. The proposed method dynamically regulates parameter updates to mitigate catastrophic forgetting while preserving learning flexibility. Extensive experiments with MuJoCo continuous control benchmarks demonstrate that MEWC-RL++ outperforms traditional regularization-based approaches and achieves performance comparable to replay-based methods without requiring data storage. Theoretical analysis further shows that adaptive regularization bounds parameter drift during conflicting tasks and maintains constant memory complexity across long task sequences. The proposed framework is particularly suitable for automation systems. That operating in nonstationary environments, where memory efficiency, scalability and long-term stability are essential. Despite its advantages, several directions remain open for future research. First, extending the framework to task-agnostic continual learning, where task boundaries are unknown, would enhance real-world applicability. Secondly, integrating representation-level adaptation. Dynamic network expansion or modular architectures, could further improve scalability. Thirdly, exploring more advanced similarity metrics, beyond cosine alignment enhance robustness in noisy, high dimensional settings. Finally, applying MEWC-RL++ to large-scale robotic platforms and multi-agent systems represents a promising direction for practical deployment.

We believe that adaptive consolidation mechanisms, the one proposed in this work, provide a principled path toward lifelong reinforcement learning in complex automation environments.

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